

20a) $\dot{x} + 7x = 5\cos(2t)$

$$\mathcal{L}\{\dot{x}\} + \mathcal{L}\{7x\} = \mathcal{L}\{5\cos(2t)\}$$

$$sF(s) + 7F(s) = \frac{5s}{s^2 + 4}$$

$$F(s)(s+7) = \frac{5s}{s^2 + 4} \rightarrow F(s) = \frac{5s}{(s+7)(s^2+4)} = \frac{5s}{(s+7)(s+2i)(s-2i)}$$

$$\frac{5s}{(s+7)(s+2i)(s-2i)} = \frac{A}{s+7} + \frac{B}{s+2i} + \frac{C}{s-2i}$$

$$x(t) = Ae^{-7t} + Be^{-2it} + Ce^{2it}$$

$$x(t) = Ae^{-7t} + B\cos(2t) - iB\sin(2t) + C\cos(2t) + iC\sin(2t)$$

$$x(t) = Ae^{-7t} + (B+C)\cos(2t) + (C-B)i\sin(2t)$$

$$5s = A(s-2i)(s+2i) + B(s+7)(s-2i) + C(s+7)(s+2i)$$

$$s = -7: -35 = 53A \quad A = -35/53$$

$$s = -2i: -10i = B(7-2i)(-4i) \Rightarrow B = \frac{5i}{4+14i} = \frac{10i+35}{106}$$

$$s = 2i: 10i = C(7+2i)(4i) \Rightarrow C = \frac{5i}{14i-4} = \frac{-10i+35}{106}$$

$$x(t) = -35/53 e^{-7t} + \left(\frac{10i+35}{106} + \frac{-10i+35}{106}\right)\cos(2t) + \left(\frac{-10i+35}{106} - \frac{10i+35}{106}\right)i\sin(2t)$$

$$x(t) = -35/53 e^{-7t} + 35/53 \cos(2t) + \left(-\frac{10i}{53}\right)i\sin(2t)$$

$$x(t) = -35/53 e^{-7t} + 35/53 \cos(2t) + 10/53 \sin(2t)$$

20b) $\ddot{x} + 6\dot{x} + 8x = 5\sin(3t)$

$$\mathcal{L}\{\ddot{x}\} + \mathcal{L}\{6\dot{x}\} + \mathcal{L}\{8x\} = \mathcal{L}\{5\sin(3t)\}$$

$$s^2 F(s) + 6s F(s) + 8F(s) = \frac{15}{s^2 + 9}$$

$$F(s)(s^2 + 6s + 8) = \frac{15}{s^2 + 9} \rightarrow F(s) = \frac{15}{(s^2 + 9)(s^2 + 6s + 8)}$$

$$F(s) = \frac{15}{(s^2 + 9)(s+4)(s+2)}$$

$$\frac{15}{(s^2 + 9)(s+4)(s+2)} = \frac{A}{s+2} + \frac{B}{s+4} + \frac{Cs}{s^2 + 9} + \frac{D}{s^2 + 9}$$

$$15 = A(s^2 + 9)(s+4) + B(s^2 + 9)(s+2) + Cs(s+4)(s+2) + D(s+4)(s+2)$$

$$15 = A(s^3 + 4s^2 + 9s + 36) + B(s^3 + 2s^2 + 9s + 18) + Cs(s^2 + 6s + 8) + D(s^2 + 6s + 8)$$

$$15 = s^3(A+B+C) + s^2(4A+2B+6C+D) + s(9A+9B+8C+6D) + (36A+18B+8D)$$

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 4 & 2 & 6 & 1 \\ 9 & 9 & 8 & 6 \\ 36 & 18 & 0 & 8 \end{bmatrix} \begin{Bmatrix} A \\ B \\ C \\ D \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 15 \end{Bmatrix} \quad A = 15/26 \quad B = -3/10 \quad C = -18/65 \quad D = -3/65$$

$$\frac{15/26}{s+2} + \frac{-3/10}{s+4} + \frac{-18/65 s}{s^2 + 9} + \frac{-1/6 s(3)}{s^2 + 9} = F(s)$$

$$\mathcal{L}^{-1}\{F(s)\} = 15/26 e^{-2t} - 3/10 e^{-4t} - 18/65 \cos(3t) - 1/65 \sin(3t)$$

20c) $\ddot{x} + 8\dot{x} + 25x = 10u(t)$
 $\mathcal{L}\{\ddot{x}\} + \mathcal{L}\{8\dot{x}\} + \mathcal{L}\{25x\} = \mathcal{L}\{10u(t)\}$

Roots = $0, -4 \pm 3i$

$$s^2 F(s) + 8s F(s) + 25 F(s) = \frac{10}{s} \rightarrow F(s) = \frac{10}{s(s^2 + 8s + 25)}$$

$$\frac{10}{s(s^2 + 8s + 25)} = \frac{A}{s} + \frac{Bs}{s^2 + 8s + 25} + \frac{C}{s^2 + 8s + 25}$$

$$10 = s^2(A+B) + s(8A+C) + 25A$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 8 & 0 & 1 \\ 25 & 0 & 0 \end{bmatrix} \begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 10 \end{bmatrix} \quad A = 2/5 \quad B = -2/5 \quad C = -16/5$$

$$F(s) = \frac{2/5}{s} + \frac{-2/5 s}{s^2 + 8s + 25} + \frac{-16/5}{s^2 + 8s + 25} = \frac{2}{5} \left[\frac{1}{s} + \frac{-s}{s^2 + 8s + 25} + \frac{-8}{s^2 + 8s + 25} \right]$$

$$F(s) = \frac{2/5}{s} + \frac{-2/5 s}{(s+4)^2 + 9} + \frac{-16/5}{(s+4)^2 + 9}$$

$$F(s) = \frac{2/5}{s} + \frac{-2/5(s+4)}{(s+4)^2 + 9} + \frac{1 \cdot 2/5 \cdot 4 \cdot 3}{3(s+4)^2 + 9} + \frac{1}{3} \frac{-16/5 \cdot 3}{(s+4)^2 + 9}$$

$$\mathcal{L}^{-1}\{F(s)\} = 2/5 - 2/5 e^{-4t} \cos(3t) + 8/15 e^{-4t} \sin(3t) - \frac{16}{15} e^{-4t} \sin(3t)$$

$$x(t) = 2/5 - e^{-4t} \left[2/5 \cos(3t) - 8/15 \sin(3t) \right]$$

2(a) $\ddot{x} + 2\dot{x} + 2x = \sin(2t)$

$$\mathcal{L}\{\ddot{x} + 2\dot{x} + 2x\} = \mathcal{L}\{\sin(2t)\}$$

$$s^2 F(s) + 2s F(s) + 2F(s) = \frac{2}{s^2 + 4}$$

$$F(s)(s^2 + 2s + 2) = \frac{2}{s^2 + 4}$$

$$F(s) = \frac{2}{(s^2 + 4)(s^2 + 2s + 2)} = \frac{As}{s^2 + 4} + \frac{B}{s^2 + 4} + \frac{Cs}{s^2 + 2s + 2} + \frac{D}{s^2 + 2s + 2}$$

$$2 = As(s^2 + 2s + 2) + B(s^2 + 2s + 2) + Cs(s^2 + 4) + D(s^2 + 4)$$

$$2 = s^3(A + C) + s^2(2A + B + D) + s(2A + 2B + 4C) + (2B + 4D)$$

$$\begin{bmatrix} 1 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 \\ 2 & 2 & 4 & 0 \\ 0 & 2 & 0 & 4 \end{bmatrix} \begin{Bmatrix} A \\ B \\ C \\ D \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ 2 \end{Bmatrix}$$

$$A = -1/5$$

$$B = -1/5$$

$$C = 1/5$$

$$D = 3/5$$

$$F(s) = \frac{As}{s^2 + 4} + \frac{B}{s^2 + 4} + \frac{Cs}{(s+1)^2 + 1} + \frac{D}{(s+1)^2 + 1}$$

$$= \frac{As}{s^2 + 4} + \frac{B}{s^2 + 4} + \frac{C(s+1)}{(s+1)^2 + 1} + \frac{-C}{(s+1)^2 + 1} + \frac{D}{(s+1)^2 + 1}$$

$$\mathcal{L}^{-1}\{F(s)\} = A \cos(2t) + \frac{B}{2} \sin(2t) + C e^{-t} \cos(t) - C e^{-t} \sin(t) + D e^{-t} \sin(t)$$

$$x(t) = \left\{ A \cos(2t) + \frac{B}{2} \sin(2t) + e^{-t} (C \cos(t) + (D - C) \sin(t)) \right\}$$

$$x(t) = -1/5 \cos(2t) - 1/10 \sin(2t) + e^{-t} (1/5 \cos(t) + 2/5 \sin(t))$$

$$21b) \ddot{x} + 2\dot{x} + x = 5e^{-2t} + t$$

$$\mathcal{L}\{\ddot{x} + 2\dot{x} + x\} = \mathcal{L}\{5e^{-2t} + t\}$$

$$s^2 F(s) + 2s F(s) + F(s) = \frac{5}{s+2} + \frac{1}{s^2}$$

$$F(s)(s^2 + 2s + 1) = \frac{5}{s+2} + \frac{1}{s^2}$$

$$F(s)(s+1)^2 = \frac{5}{s+2} + \frac{1}{s^2}$$

$$F(s) = \frac{5}{(s+1)^2(s+2)} + \frac{1}{s^2(s+1)^2} = \frac{5s^2 + s + 2}{(s+1)^2(s+2)s^2}$$

$$\frac{5s^2 + s + 2}{(s+1)^2(s+2)s^2} = \frac{A}{(s+1)^2} + \frac{B}{s+1} + \frac{C}{s^2} + \frac{D}{s} + \frac{E}{s+2}$$

$$5s^2 + s + 2 = A(s^3 + 2s^2) + B(s^4 + 3s^3 + 2s^2) + C(s^3 + 4s^2 + 5s + 2) + D(s^4 + 4s^3 + 5s^2 + 2s) + E(s^4 + 2s^3 + s^2)$$

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 1 \\ 1 & 3 & 1 & 4 & 2 \\ 2 & 2 & 4 & 5 & 1 \\ 0 & 0 & 5 & 2 & 0 \\ 0 & 0 & 2 & 0 & 0 \end{bmatrix} \begin{Bmatrix} A \\ B \\ C \\ D \\ E \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 5 \\ 1 \\ 2 \end{Bmatrix} \quad \begin{array}{l} A = 6 \\ B = -3 \\ C = 1 \\ D = -2 \\ E = 5 \end{array}$$

$$\mathcal{L}^{-1}\{F(s)\} = Ate^{-t} + Be^{-t} + Ct + D + Ee^{-2t}$$

$$x(t) = 6te^{-t} - 3e^{-t} + t - 2 + 5e^{-2t}$$

$$2c) \ddot{x} + 4x = t^2$$

$$\mathcal{L}\{\ddot{x} + 4x\} = \mathcal{L}\{t^2\}$$

$$s^2 F(s) + 4F(s) = \frac{2}{s^3}$$

$$F(s)(s^2 + 4) = \frac{2}{s^3} \rightarrow F(s) = \frac{2}{s^3(s^2 + 4)}$$

$$\frac{2}{s^3(s^2 + 4)} = \frac{A}{s^3} + \frac{B}{s^2} + \frac{C}{s} + \frac{D}{(s^2 + 4)}$$

$$2 = A(s^2 + 4) + Bs(s^2 + 4) + Cs^2(s^2 + 4) + D(s^3)$$

$$2 = s^4(C) + s^3(B + D) + s^2(A + 4C) + s(4B) + 4A$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & A & 0 \\ 0 & 1 & 0 & 1 & B & 0 \\ 0 & 4 & 0 & 0 & C & 0 \\ 4 & 0 & 0 & 0 & D & 2 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 4 & 0 \\ 0 & 4 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 4 & 0 \\ 0 & 4 & 0 & 0 \\ 4 & 0 & 0 & 0 \end{bmatrix}$$

$$A = \frac{1}{2} \quad s = 0$$

$$s = 4i$$

$$-64D = 2$$

$$D = -\frac{1}{32}$$

$$s = -4i$$

$$x(t) = \frac{1}{2}t^2 + \frac{1}{2}D \sin(2t)$$

$$x(t) = \frac{1}{2}t^2 - \frac{1}{64} \sin(2t)$$

$$x(t) = \frac{1}{2}t^2 + \frac{1}{2}D \sin(2t)$$

$$x(t) = \frac{1}{2}t^2 - \frac{1}{64} \sin(2t)$$